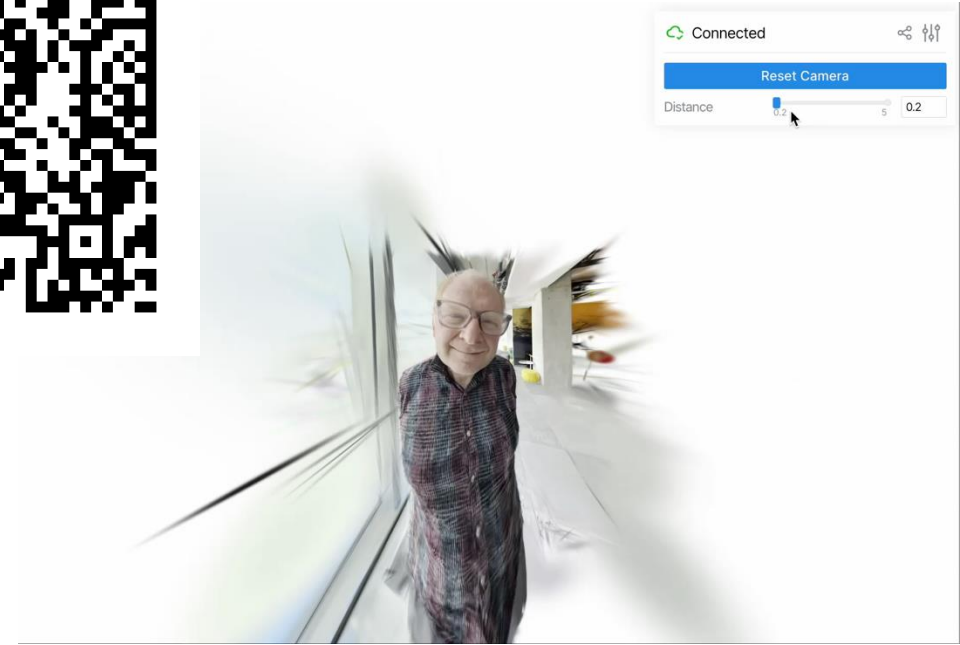
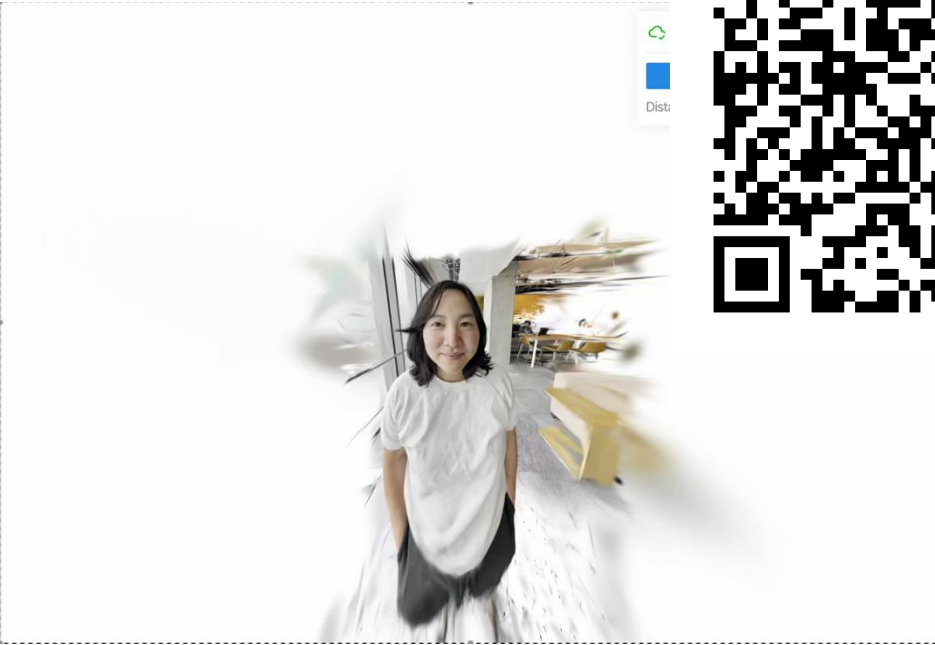


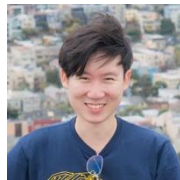


Feedback on Discussion #1



CS180/280A Discussion #2

Hi 🖐️ I'm Konpat Preechakul



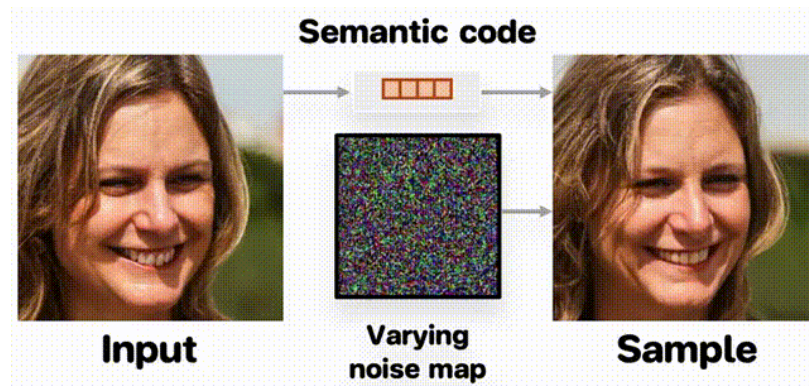
Me: **Konpat** Preechakul

3rd year 🙌 student.

Scene understanding

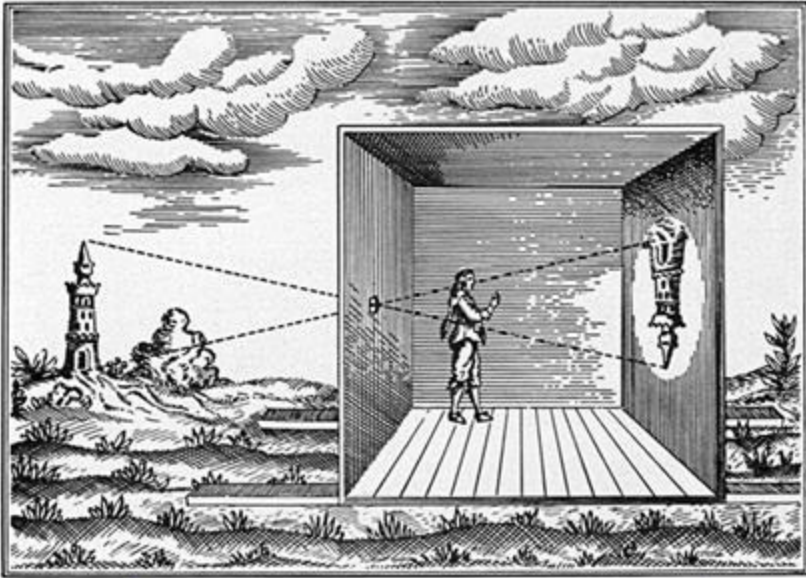


Diffusion models & Representation learning



Today: image formation

Relationship between points in 3D and 2D



Move to a person next to you,
And introduce yourself 😊 (1 min)

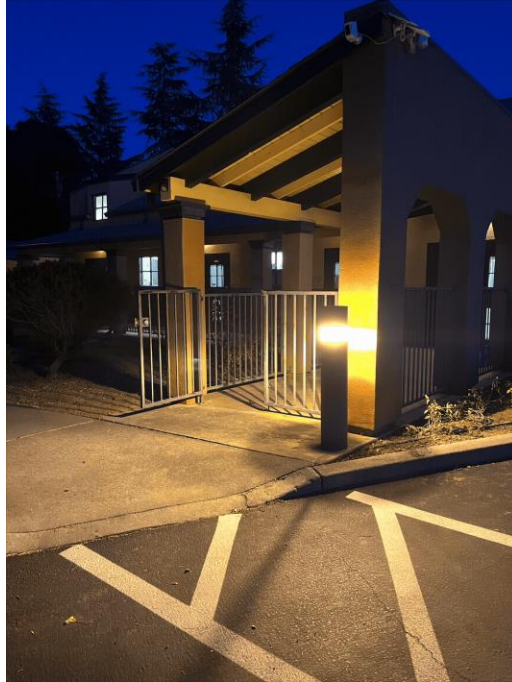
“Hi what’s your name?”

“What’s your favorite fruit?”

“Where are you heading to after this?”

Let’s work in pairs!

Project#0 Recap...



Are the two images taken by
only zooming
(and how do you know?)

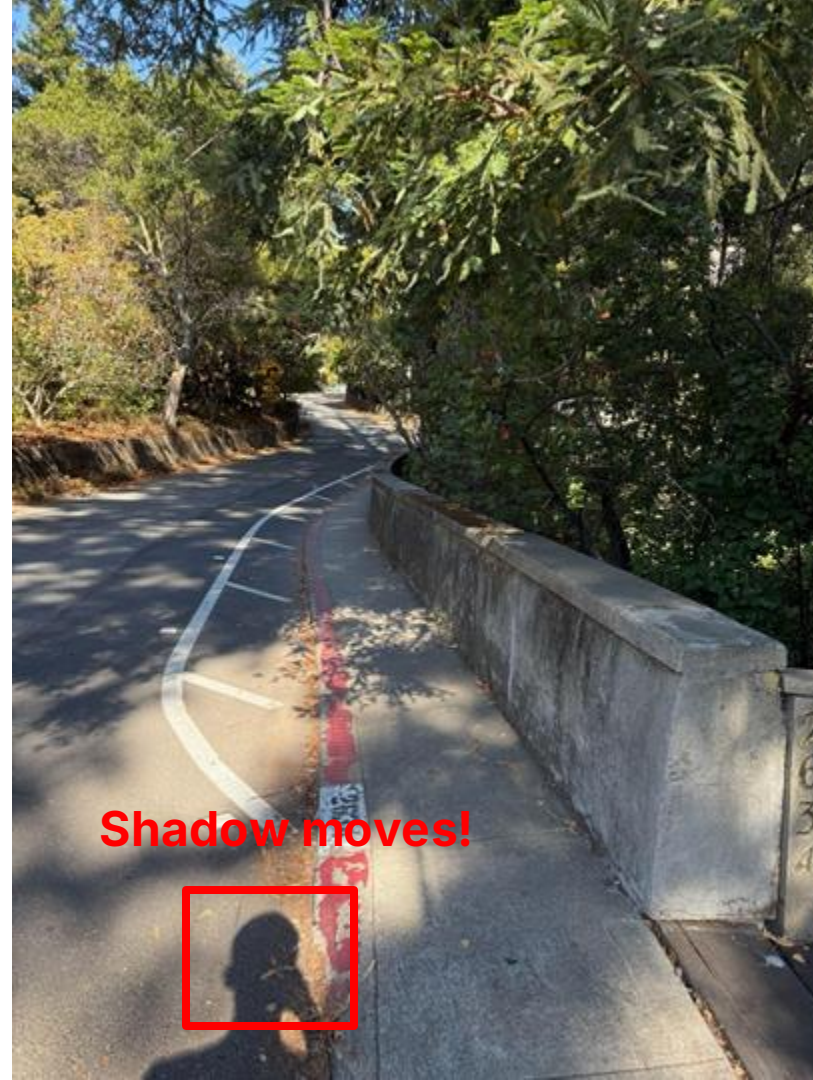
Discuss in pairs (1 min) and come up with a proof!



12
MM



24
MM



One more...
Is it just zoom?

Discuss in pairs (3 mins) and come up with a proof!



48
MM



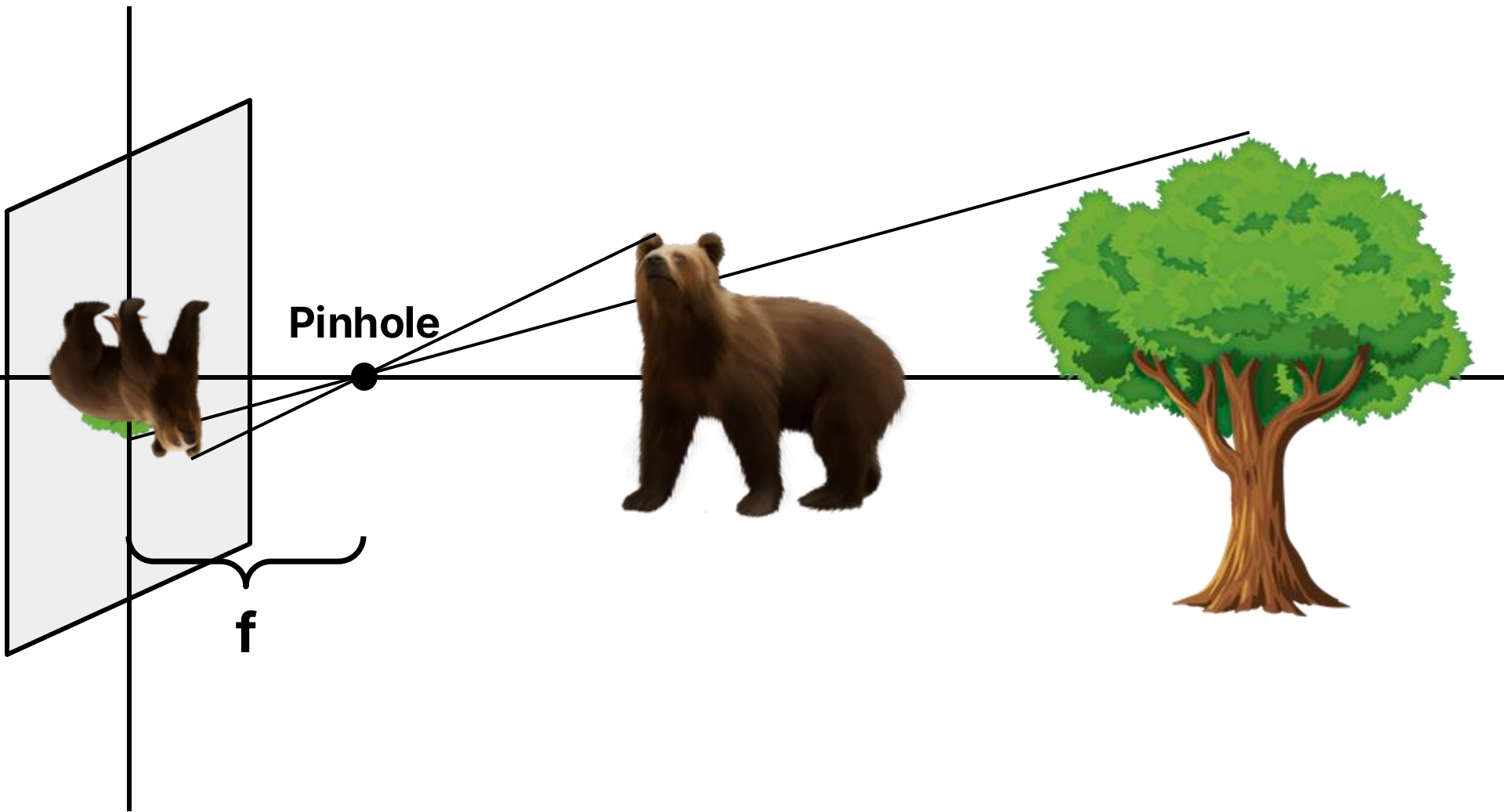
120
MM

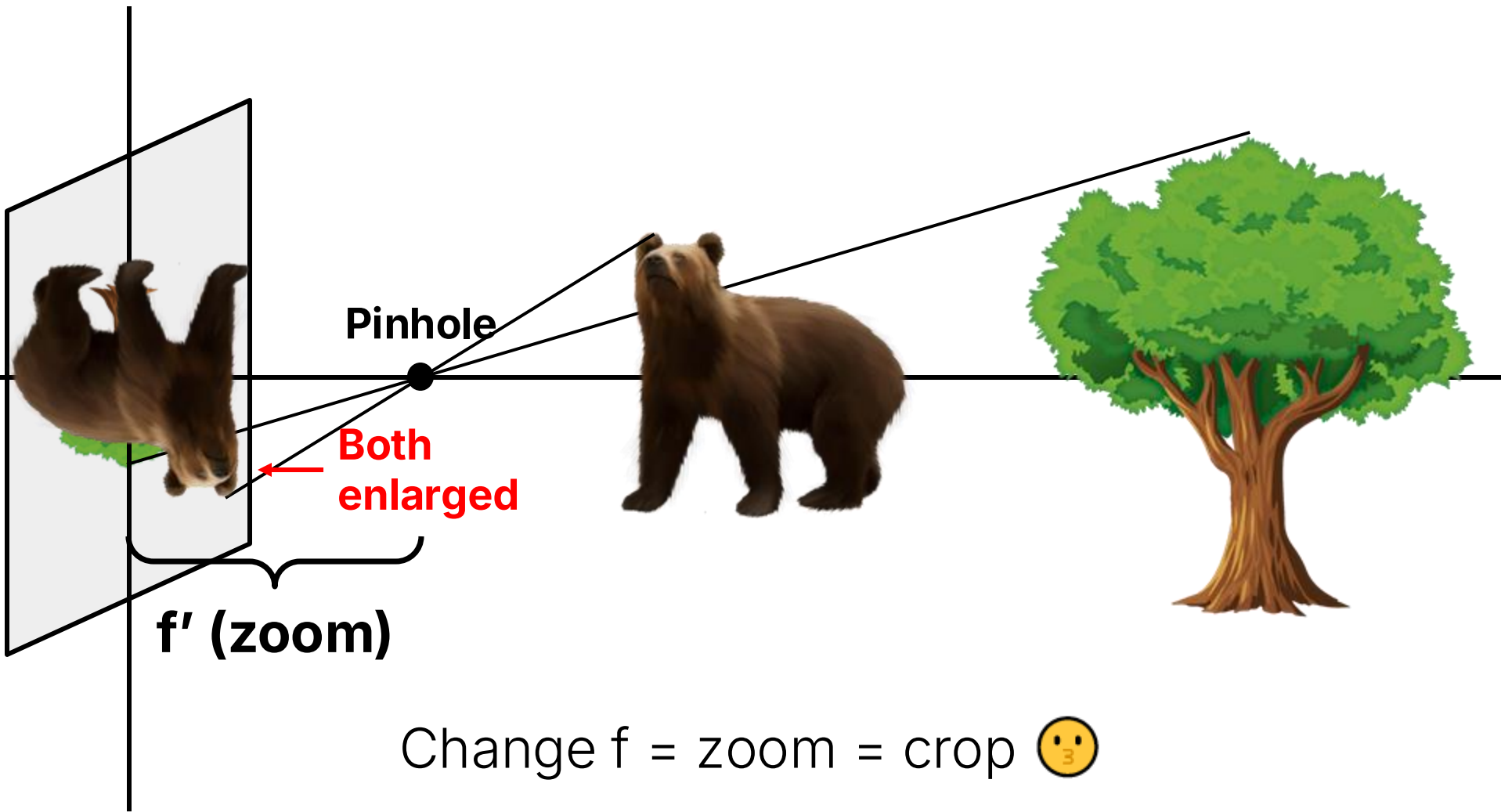


So compressed!

Focal length = zoom = crop*

Perspectively speaking, crop = zoom, but with lower resolution





When Zoom $\neq$ Crop (Lens flaws; pinhole is fine)

Chromatic Aberration

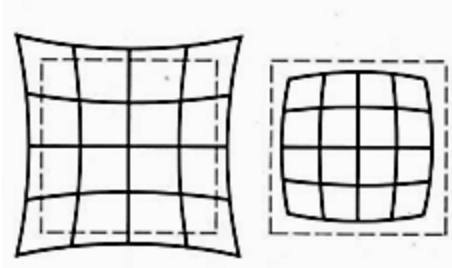
Near Lens Center



Near Lens Outer Edge



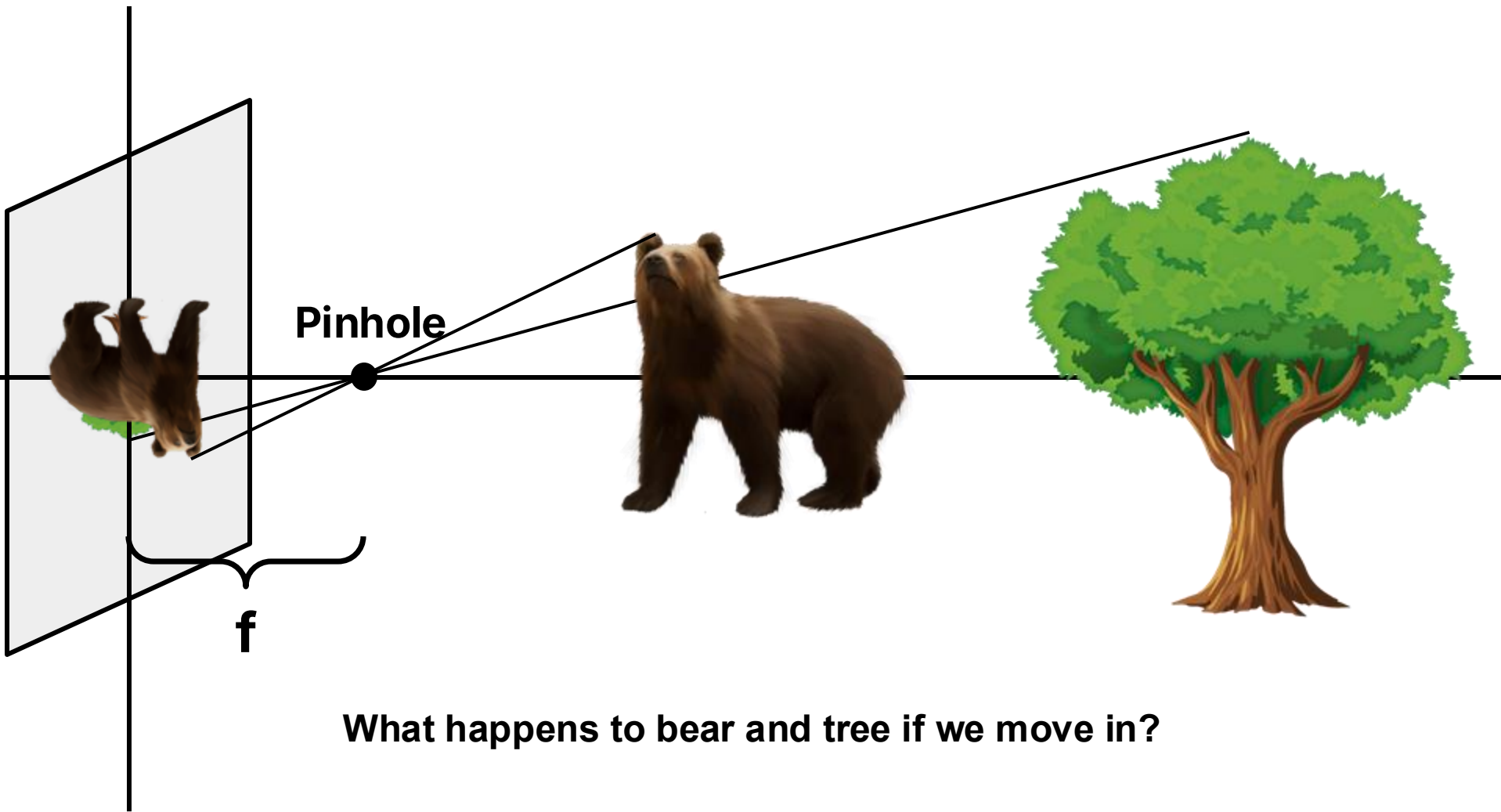
Radial Distortion

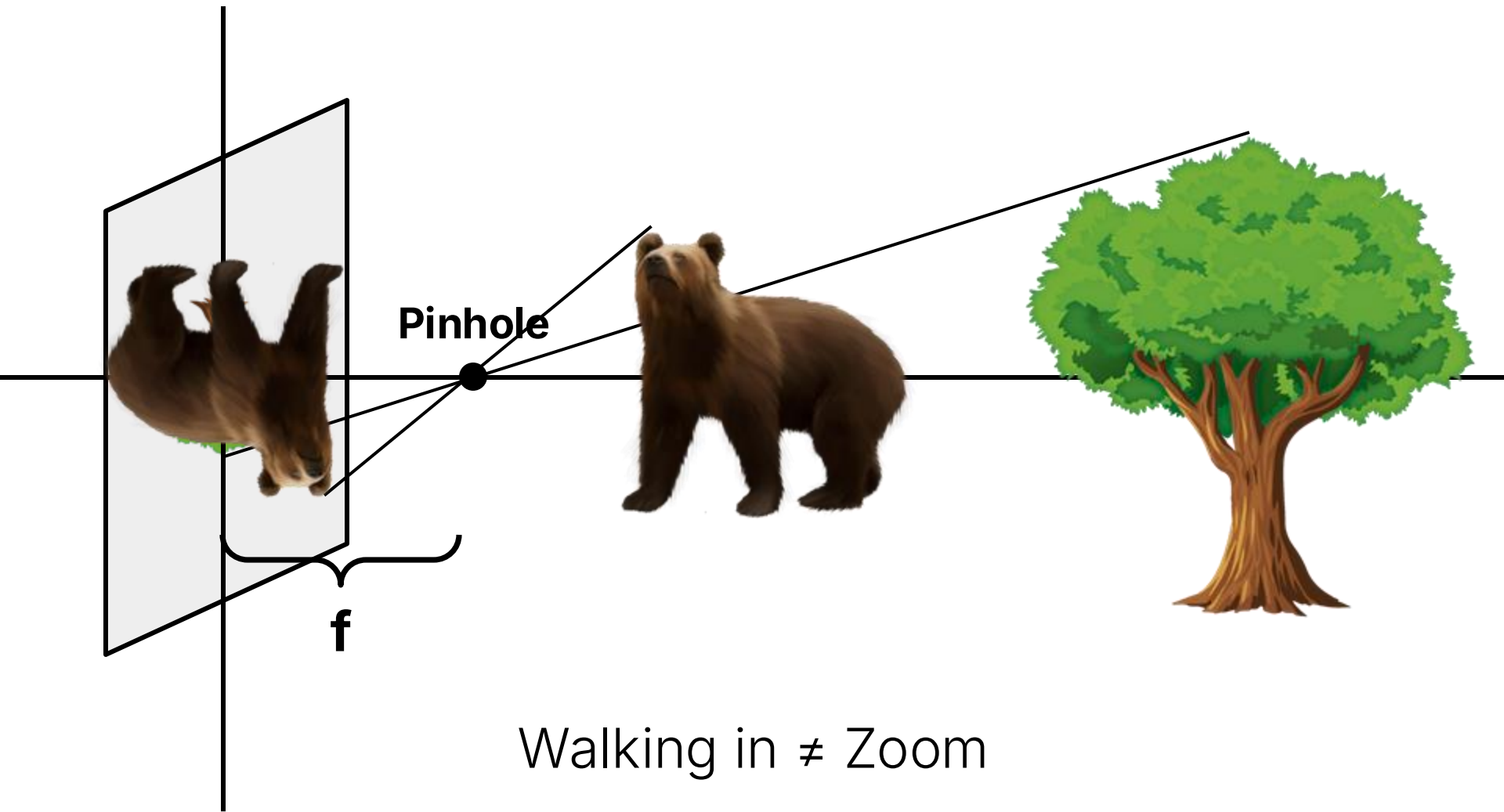


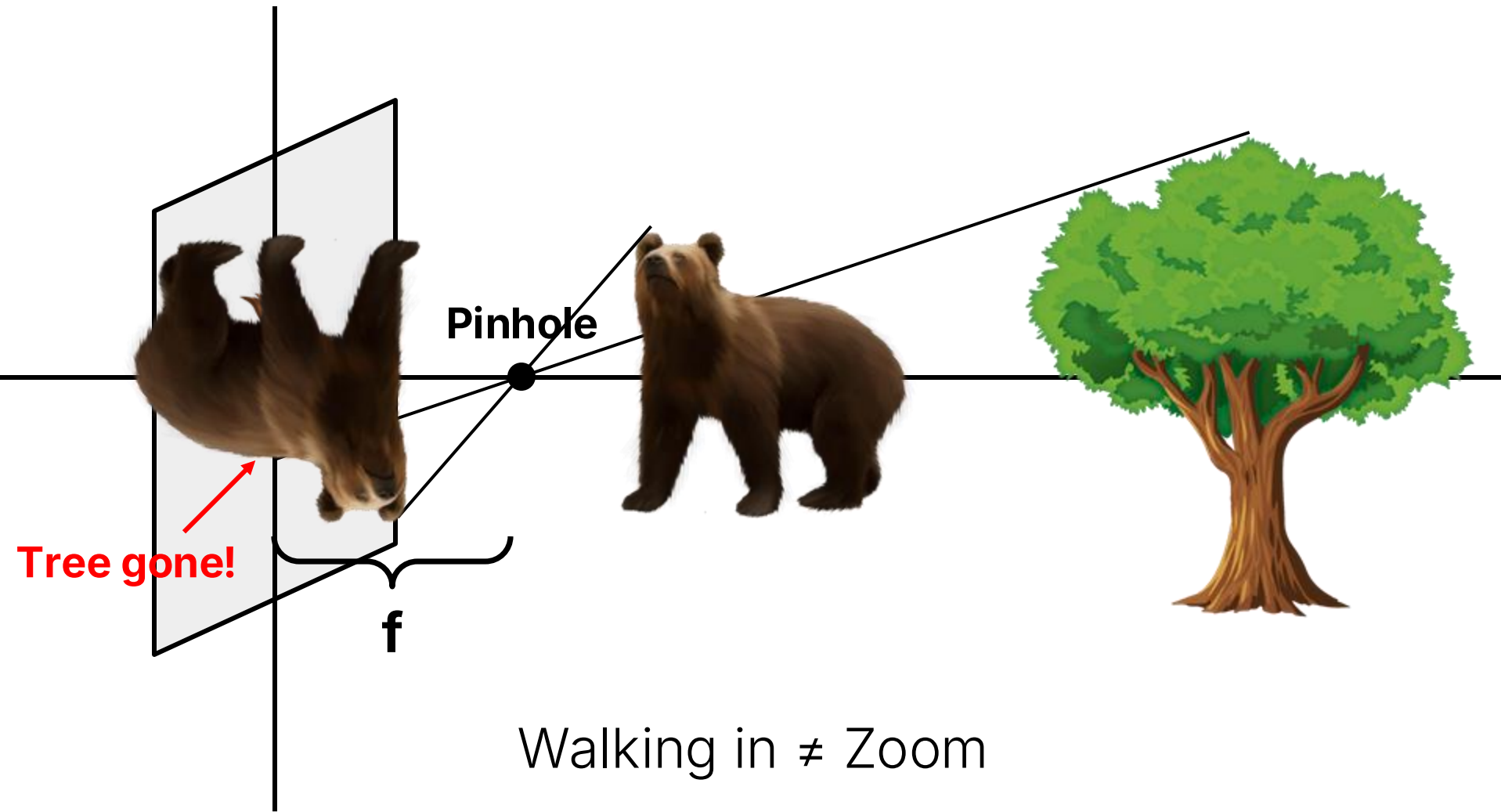
Pin cushion Barrel



(Focal length = zoom = crop)
≠ Move

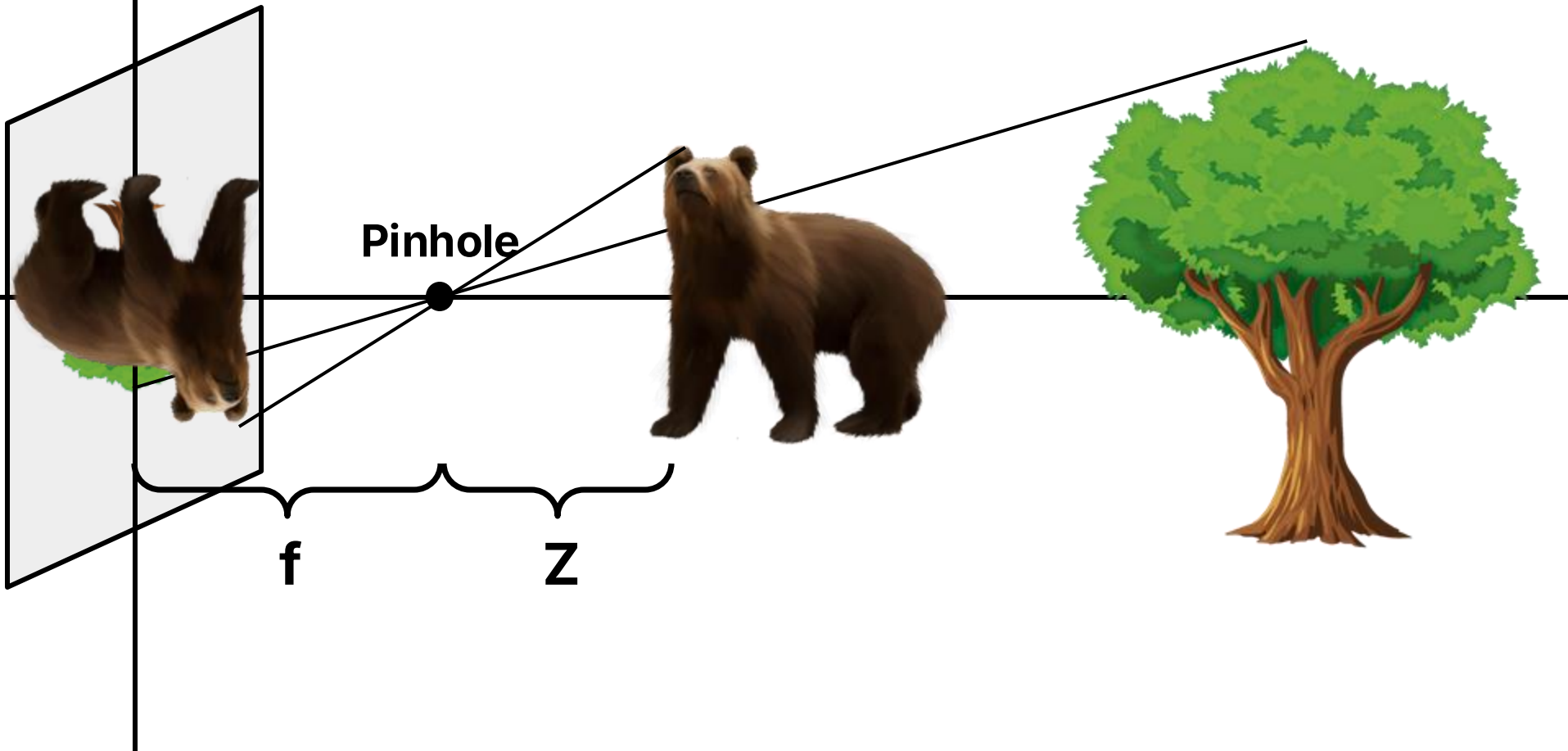




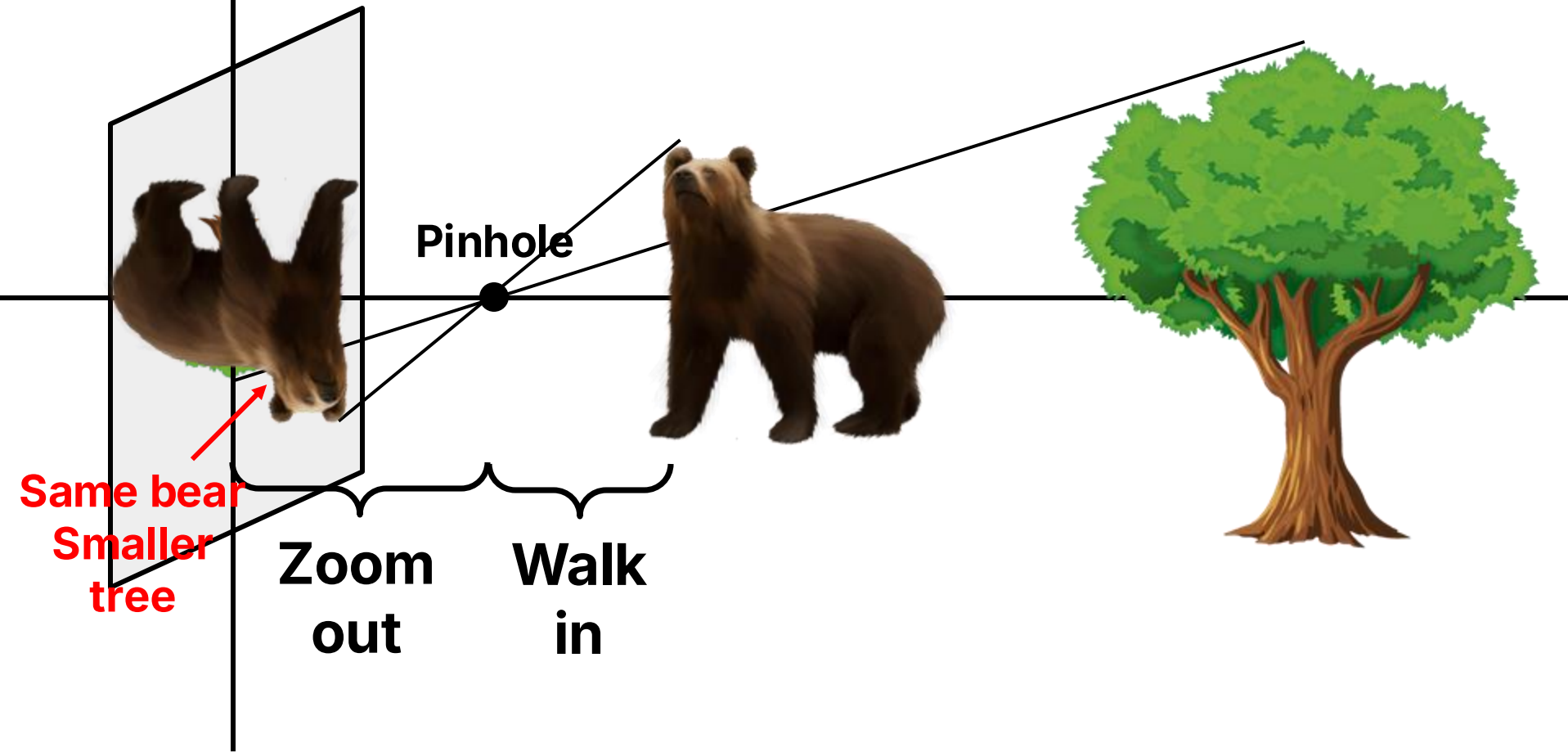


Dolly zoom: walking while
zooming

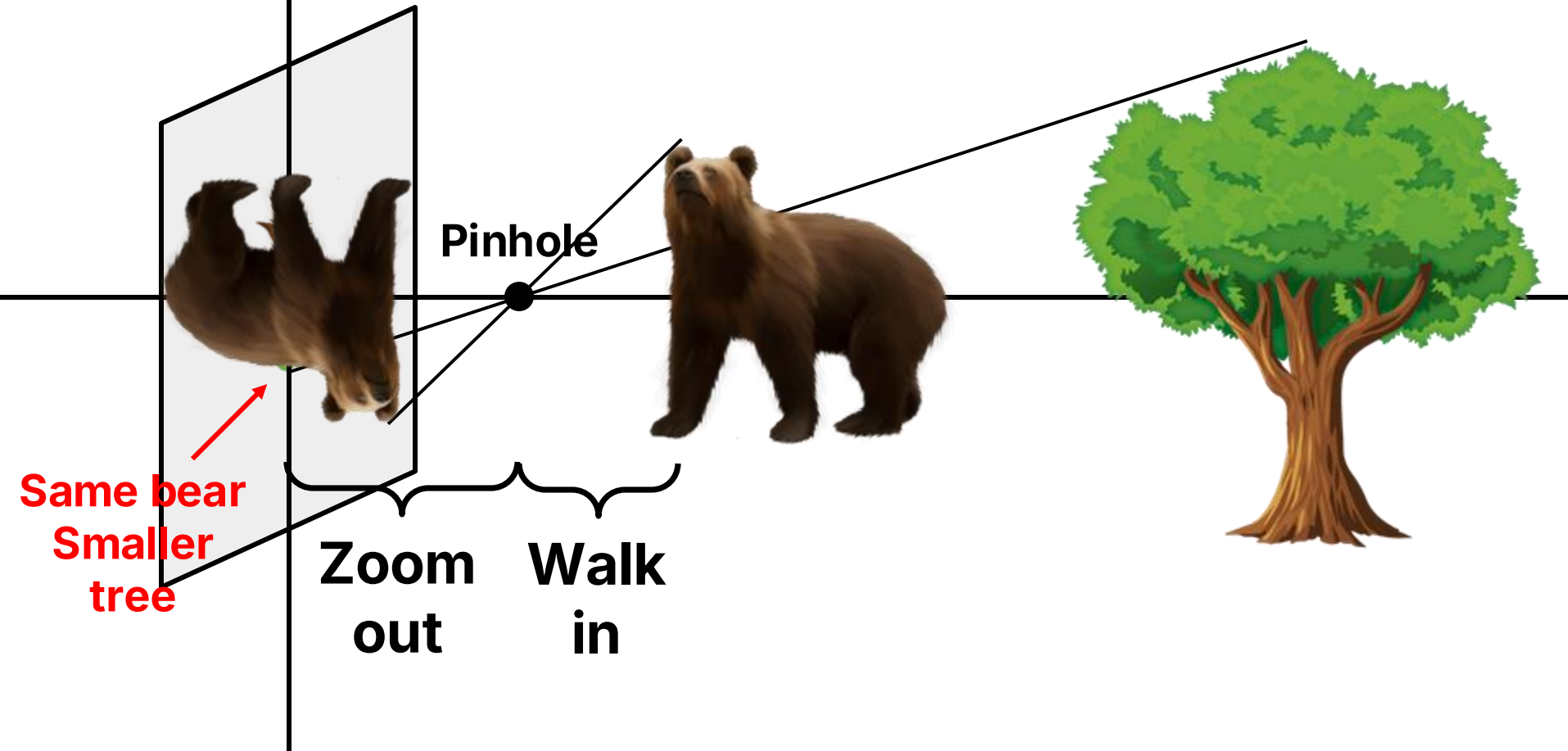
Dolly zoom: walking while zooming



Dolly zoom: walking while zooming

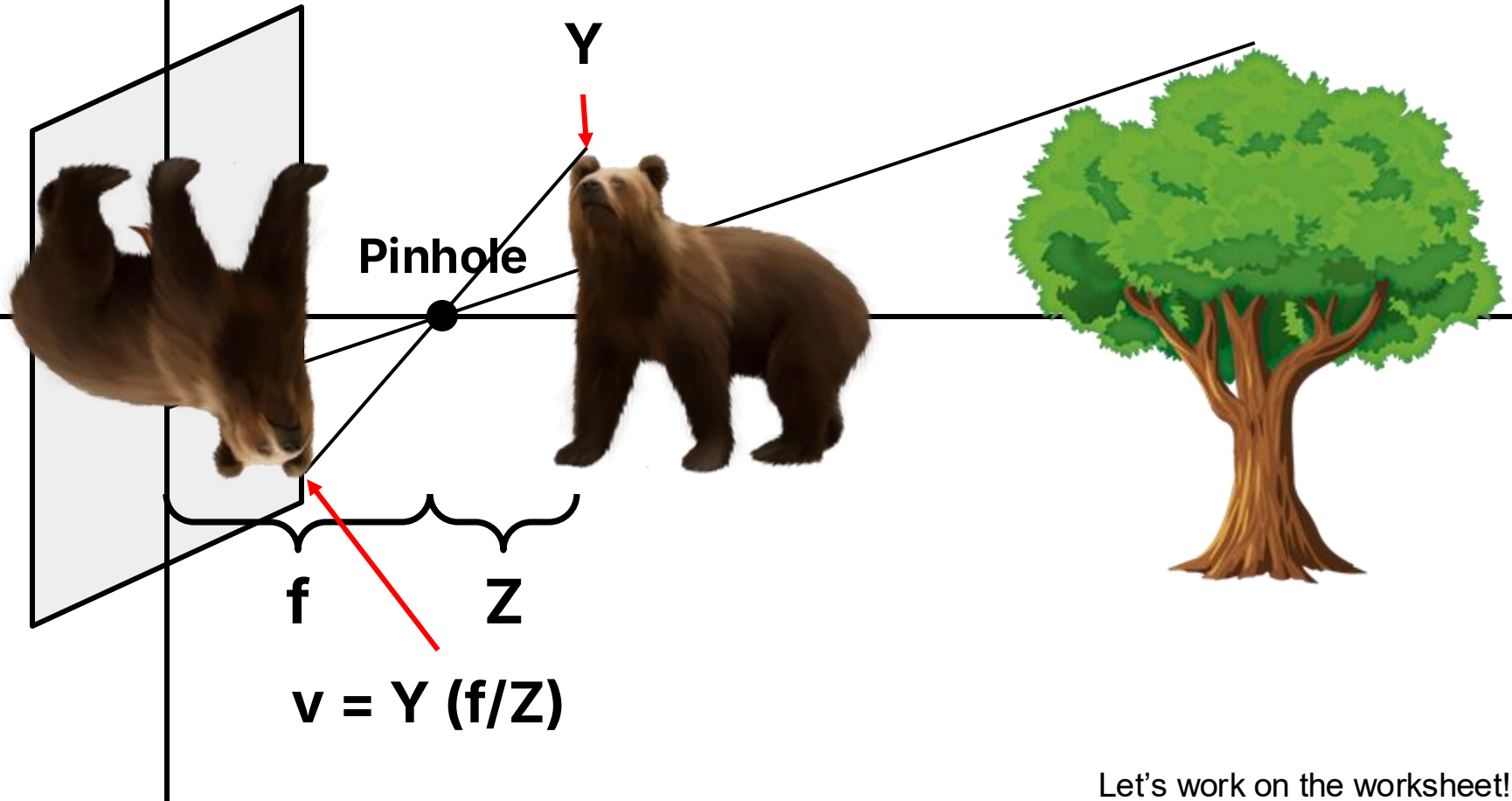


Dolly zoom: walking while zooming



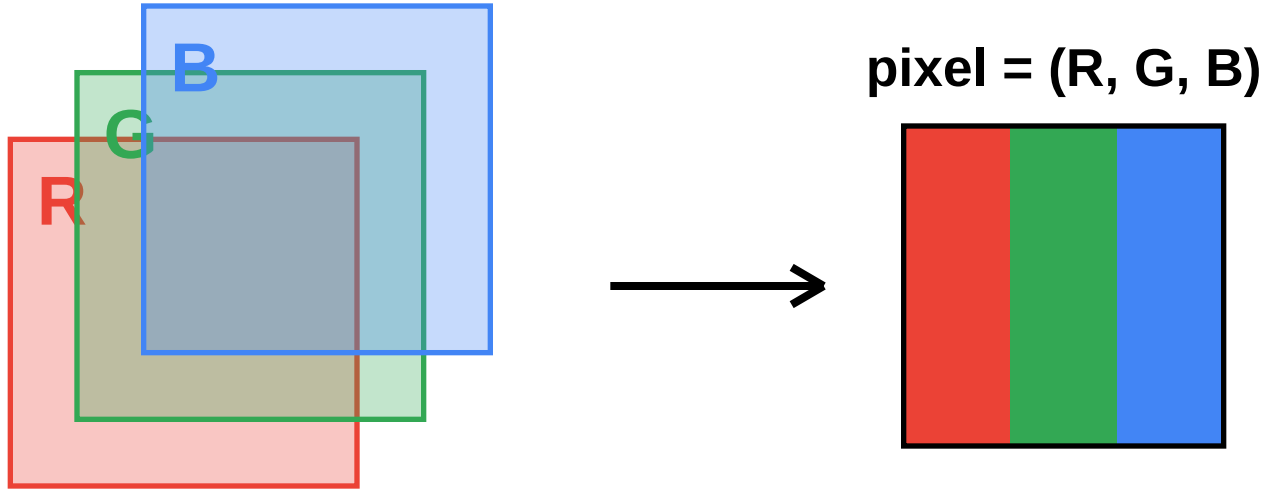
Pinhole camera

In a nutshell



Let's work on the worksheet!

Color: one image = three grayscale channels



Each channel alone is just a grayscale image — an intensity map for one primary

Which channel is bright?

Discuss in pairs (2 mins)!

Red stop sign

→ **bright in R? G? B?**

Green grass

White clouds

Bonus: grayscale is $Y = 0.299R + 0.587G + 0.114B$.

Why not just average the three channels?